

Rukesh Prajapati

Personal Information

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Summary

Ph.D. student in Computer Engineering at West Virginia University with research and industry experience in computer vision, robotics, autonomous systems, and deep learning. Hands-on background in perception for robotics and self-driving applications, including bird's-eye-view transformation, real-time vision pipelines, tracking, and ML-based scene understanding. Experience with vision-language models (VLMs) and vision-language-action (VLA) frameworks for decision-making, along with full-stack development for robotic systems integrating perception, localization, and planning. Published in peer-reviewed journals and IEEE conferences, with additional industry experience building computer vision systems for sports analytics and intelligent automation.

Education

Ph.D. in Computer Engineering West Virginia University, USA	2023 – Present
M.E. in Information and Communication Engineering Chosun University, South Korea	2020 – 2022
B.E. in Electronics and Communication Engineering Khwopa Engineering College, Nepal	2012 – 2017

Technical Skills

Programming: Python, C++, C
Machine Learning: PyTorch, TensorFlow
Computer Vision: OpenCV, Object Detection, Multi-Object Tracking, Image Segmentation
Robotics & Autonomous Systems: ROS, Perception for Autonomous Systems, SLAM (LiDAR-based), GNSS/GPS Localization, Trajectory Planning, VLMs, VLA Models
Simulation: NVIDIA Isaac Sim, CARLA, Webots, RoadRunner
Embedded Systems: NVIDIA Jetson, Raspberry Pi, Arduino

Experience

Graduate Research Assistant West Virginia University, USA	Aug 2023 – Present
- Develop AI-driven perception and decision-making algorithms for autonomous vehicle applications as part of the EcoCAR Innovation Track. - Developed perception and decision-making frameworks for autonomous robotic systems using deep learning and computer vision. - Designed and trained robotic platforms in simulation environments (CARLA, NVIDIA Isaac Sim) using deep reinforcement learning (DRL). - Worked on VLMs and VLA approaches for high-level decision-making in robotics. - Built integrated pipelines combining perception, localization, and trajectory planning for autonomous navigation.	
Research Assistant Chosun University, South Korea	Sep 2020 – Jun 2023
- Designed deep learning models for Alzheimer's disease classification. - Developed and optimized neural networks for MRI segmentation. - Published results in journals and international conferences.	
Computer Vision Engineer Newrun Technology Pvt. Ltd., Nepal	Jul 2019 – Aug 2020
Built a computer vision pipeline for sports video analysis and player recommendation.	

- Implemented object detection, face recognition, motion tracking, and ball tracking.

Selected Technical Projects

Vision-Language Decision Making for SPARK Robot Platform (Indro Robotics)

- Developed VLM-based decision-making framework for robotic task execution using multimodal perception.

Sim-to-Real Zero-Shot Learning for Autonomous Driving

- Developed zero-shot sim-to-real transfer approach for autonomous driving policies using simulation environments.

Bird's-Eye-View Transformation for Autonomous Driving

- Developed spatial transformer-based perspective mapping for road-scene understanding.

Sports Video Analytics System

- Built computer vision system for player and ball tracking using deep learning.

Publications

Journal Articles

- [1] **R. Prajapati** and A. S. El-Wakeel, "Masseg: Motion aware semantic segmentation for autonomous vehicles," Manuscript under review at IEEE Sensors, 2026.
- [2] **R. Prajapati** and A. S. El-Wakeel, "See first, think if needed: Embodied conditional VLA for autonomous mobile robot navigation," Manuscript under review, 2026.
- [3] **R. Prajapati** and A. S. El-Wakeel, "Sim2real dynamic road intersection monitoring via adaptive vehicle bev transformation model for the internet of vehicles," *IEEE Internet of Things Journal*, vol. 13, no. 17, pp. 39 683–39 701, 2026. doi: 10.1109/JIOT.2026.3709220.
- [4] A. Devkota, **R. Prajapati**, A. El-Wakeel, D. Adjeroh, B. Patel, and P. Gyawali, "Ai analysis for ejection fraction estimation from 12-lead ecg," *Scientific Reports*, vol. 15, no. 1, p. 13 502, 2025.
- [5] **R. Prajapati**, G.-R. Kwon, et al., "A binary classifier using fully connected neural network for alzheimer's disease classification," *Journal of Multimedia Information System*, vol. 9, no. 1, pp. 21–32, 2022.
- [6] **R. Prajapati** and G.-R. Kwon, "Dissimilarity corrective generative adversarial network for brain image segmentation," *Applied Sciences*, vol. 12, no. 24, p. 12 944, 2022.
- [7] **R. Prajapati** and G.-R. Kwon, "Sip-unet: Sequential inputs parallel unet architecture for segmentation of brain tissues from magnetic resonance images," *Mathematics*, vol. 10, no. 15, 2022, ISSN: 2227-7390. doi: 10.3390/math10152755. [Online]. Available: <https://www.mdpi.com/2227-7390/10/15/2755>.

Conference Papers

- [8] D. Babalola, **R. Prajapati**, and A. S. El-Wakeel, "Real-time analysis of yolov12 and rt-detr for object detection in autonomous driving," in *2026 International Conference on Smart Mobility (SM)*, 2026, pp. 1–6. doi: 10.1109/SM69703.2026.11614124.
- [9] **R. Prajapati** and A. S. El-Wakeel, "Camera perspective transformation to bird's eye view via spatial transformer model for road intersection monitoring," in *2024 IEEE International Conference on Smart Mobility (SM)*, 2024, pp. 135–140. doi: 10.1109/SM63044.2024.10733499.
- [10] **R. Prajapati** and A. S. El-Wakeel, "Cloud-based federated learning framework for mri segmentation," in *ICC 2024 - IEEE International Conference on Communications*, 2024, pp. 220–225. doi: 10.1109/ICC51166.2024.10622470.
- [11] **R. Prajapati**, U. Khatri, and G. R. Kwon, "An efficient deep neural network binary classifier for alzheimer's disease classification," in *2021 International Conference on Artificial Intelligence in Information and Communication (ICAIIIC)*, 2021, pp. 231–234. doi: 10.1109/ICAIIIC51459.2021.9415212.

Honors and Awards

- Runner-up, **F1TENTH Autonomous Grand Prix**, IEEE Smart Mobility 2024
- **Dianne Dubetz Anderson Fellowship** 2024, West Virginia University
- **NSF Student Travel Grant** (IEEE ICC 2024, AraFest 2024)
- **Multiple National Robotics Competition Wins** in Nepal, (IOE Robocon, FPGA Design Competition, and Yantra Robotics Competition) (2015-2017)